

MAT41-1

Linear Algebra IV

SVD & Applications

The Singular Value Decomposition and its applications to least squares, dimensionality reduction, and numerical stability.

Albert School — 30 hours

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1 The Singular Value Decomposition

Throughout, $A \in M_{m,n}(\mathbb{R})$ is a real $m \times n$ matrix of rank r . We use the eigendecomposition and the spectral theorem (MAT21-1, MAT31-1).

Definition 1 (Singular value decomposition) A **singular value decomposition** of $A \in M_{m,n}(\mathbb{R})$ is a factorization

$$A = U\Sigma V^T,$$

where $U \in M_m(\mathbb{R})$ and $V \in M_n(\mathbb{R})$ are orthogonal and $\Sigma \in M_{m,n}(\mathbb{R})$ is diagonal (in the rectangular sense: $\Sigma_{ij} = 0$ for $i \neq j$) with diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots,$$

where $r = \text{rank}(A)$.

The σ_i are the **singular values** of A , the columns of U are the **left singular vectors**, and the columns of V are the **right singular vectors**.

Theorem 2 (Existence) Every matrix $A \in M_{m,n}(\mathbb{R})$ admits a singular value decomposition.

1.1 Computation via $A^T A$

The matrix $A^T A \in M_n(\mathbb{R})$ is symmetric and positive semi-definite, so by the spectral theorem (MAT31-1) it is orthogonally diagonalizable with non-negative eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$.

Proposition 3 (SVD from the eigendecomposition of $A^T A$) Let $A^T A = VDV^T$ with V orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. Then:

- the singular values are $\sigma_i = \sqrt{\lambda_i}$;
- the columns of V are the right singular vectors;
- for each i with $\sigma_i > 0$, the corresponding left singular vector is

$$u_i = \frac{1}{\sigma_i} A v_i;$$

- the vectors $u_1, \dots, u_r$ are orthonormal and are completed to an orthonormal basis of $\mathbb{R}^m$ to form U .

Procedure. To compute an SVD of A :

1. form $A^T A$ and compute its eigenvalues λ_i and an orthonormal basis of eigenvectors v_i (ordering $\lambda_1 \geq \dots \geq \lambda_n$);
2. set $\sigma_i = \sqrt{\lambda_i}$ and let $V = (v_1 \mid \dots \mid v_n)$;
3. for $\sigma_i > 0$, set $u_i = Av_i/\sigma_i$; complete $u_1, \dots, u_r$ to an orthonormal basis to get U ;
4. assemble Σ with the σ_i on its diagonal; then $A = U\Sigma V^T$.

2 Geometric interpretation

Writing $A = U\Sigma V^T$, the action $x \mapsto Ax$ factors as

- V^T : an orthogonal transformation (rotation/reflection) of the input;
- Σ : a non-negative scaling of the coordinate axes by the singular values;
- U : an orthogonal transformation of the output.

Thus every linear map is a **rotate–stretch–rotate**.

Proposition 4 (Image of the unit sphere) The image under A of the unit sphere $\{x \in \mathbb{R}^n : \|x\| = 1\}$ is an ellipsoid in $\mathbb{R}^m$ whose semi-axes have lengths equal to the singular values $\sigma_1, \dots, \sigma_r$, directed along the left singular vectors $u_1, \dots, u_r$.

3 Connection to the spectral theorem

Proposition 5 (SVD of a symmetric PSD matrix) If A is symmetric positive semi-definite, its spectral decomposition $A = PDP^T$ (with P orthogonal and D diagonal with non-negative entries, MAT31-1) is also a singular value decomposition: the singular values are the eigenvalues, and the left and right singular vectors coincide with the eigenvectors ($U = V = P$).

For a general symmetric matrix with a negative eigenvalue λ , the corresponding singular value is $|\lambda|$, and the sign is absorbed into the left singular vector ($u_i = -v_i$); so the SVD and the eigendecomposition coincide exactly when A is positive semi-definite.

4 Matrix norms

Definition 6 (Frobenius and spectral norms) For $A \in M_{m,n}(\mathbb{R})$ with singular values $\sigma_1 \geq \dots \geq \sigma_r$:

- the **Frobenius norm** is

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\sum_{i=1}^r \sigma_i^2};$$

- the **spectral norm** is

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\| = \sigma_1.$$

Definition 7 (Condition number) For an invertible (or full-column-rank) matrix A with smallest non-zero singular value σ_r , the **condition number** is

$$\kappa(A) = \frac{\sigma_1}{\sigma_r}.$$

A large condition number means that the matrix is **ill-conditioned**: small relative changes in the data can produce large relative changes in the solution of a linear system. One always has $\kappa(A) \geq 1$, and $\kappa(A) = 1$ for orthogonal matrices.

5 Low-rank approximation

For $k \leq r$, truncating the SVD after k terms gives

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T = U \Sigma_k V^T,$$

where Σ_k keeps only the k largest singular values. The matrix A_k has rank k .

Theorem 8 (Eckart–Young) Among all matrices of rank at most k , the truncated SVD A_k is a best approximation of A in both the spectral and Frobenius norms:

$$\begin{aligned} \min_{\text{rank}(B) \leq k} \|A - B\|_2 &= \|A - A_k\|_2 = \sigma_{k+1}, \\ \min_{\text{rank}(B) \leq k} \|A - B\|_F &= \|A - A_k\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}. \end{aligned}$$

The approximation error is therefore controlled by the discarded singular values. Choosing k trades storage (a rank- k approximation needs $k(m+n+1)$ numbers) against fidelity (the error above): this is the **compression–quality tradeoff**.

6 The pseudo-inverse

Definition 9 (Moore–Penrose pseudo-inverse) Let $A = U \Sigma V^T$. The **pseudo-inverse** of A is

$$A^+ = V \Sigma^+ U^T,$$

where $\Sigma^+ \in M_{n,m}(\mathbb{R})$ is obtained from Σ by transposing it and replacing each non-zero singular value σ_i by $1/\sigma_i$ (leaving zeros in place).

Proposition 10 (Penrose conditions) The pseudo-inverse $X = A^+$ is the unique matrix satisfying the four **Penrose conditions**:

$$AXA = A, \quad XAX = X, \quad (AX)^T = AX, \quad (XA)^T = XA.$$

Proposition 11 (Minimum-norm least-squares solution) For any b , the vector $x^* = A^+b$ is the least-squares solution of $Ax = b$ of smallest Euclidean norm: it minimizes $\|Ax - b\|$, and among all minimizers it is the one with smallest $\|x\|$.

7 Least squares

Consider an **over-determined** system $Ax = b$ with $A \in M_{m,n}(\mathbb{R})$, $m > n$, generally inconsistent. The **least-squares** problem is to minimize $\|Ax - b\|$, with **residual** $r = b - Ax$.

Theorem 12 (Normal equations and residual orthogonality) x minimizes $\|Ax - b\|$ if and only if it satisfies the **normal equations**

$$A^T Ax = A^T b, \quad \text{equivalently} \quad A^T r = 0.$$

That is, the residual $r = b - Ax$ is orthogonal to the column space of A . When A has full column rank, the solution is unique and equals $x = (A^T A)^{-1} A^T b = A^+ b$.

The residual orthogonality $A^T r = 0$ expresses that Ax is the orthogonal projection of b onto the column space of A (best approximation, MAT31-1). The reliability of the computed solution is governed by the condition number $\kappa(A)$: the larger it is, the more sensitive x is to perturbations of b .

8 Principal Component Analysis

Let $X \in M_{m,n}(\mathbb{R})$ hold m observations (rows) of n features (columns).

PCA procedure.

1. **Center** the data: subtract from each column its mean, giving $\tilde{X}$ with zero-mean columns.
2. Form the **covariance matrix**

$$C = \frac{1}{m-1} \tilde{X}^T \tilde{X} \in M_n(\mathbb{R}),$$

which is symmetric positive semi-definite.

3. **Eigendecompose** $C = V \Lambda V^T$ (spectral theorem); the eigenvectors $v_1, \dots, v_n$ are the **principal components**, ordered by decreasing eigenvalue $\lambda_1 \geq \dots \geq \lambda_n \geq 0$.

Proposition 13 (SVD link) If $\tilde{X} = U \Sigma V^T$ is an SVD of the centered data, then

$$C = \frac{1}{m-1} V \Sigma^T \Sigma V^T,$$

so the principal components are the right singular vectors v_i of $\tilde{X}$, and the eigenvalues of C are $\lambda_i = \sigma_i^2 / (m-1)$. PCA is thus computed directly from the SVD of $\tilde{X}$, without forming C .

Definition 14 (Explained variance ratio) The **explained variance ratio** of the i -th principal component is

$$\frac{\lambda_i}{\sum_{j=1}^n \lambda_j} = \frac{\sigma_i^2}{\sum_{j=1}^n \sigma_j^2}.$$

To **choose k** , one selects the smallest number of components whose cumulative explained variance ratio

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^n \lambda_j}$$

reaches a desired threshold (e.g. 0.95).

9 Limitations of PCA

PCA is appropriate only under conditions that must be checked on the data:

- **Scale sensitivity:** PCA depends on the units of the features, since the covariance matrix changes when features are rescaled; features must usually be standardized first.
- **Linearity assumption:** PCA captures only linear structure; data lying on a nonlinear manifold are not well represented by linear principal components.
- **Orthogonality constraint:** principal components are forced to be mutually orthogonal, which may not match the true structure of the data.

10 Numerical stability

Proposition 15 (Squaring of the condition number) For any matrix A , $\kappa(A^T A) = \kappa(A)^2$, because the singular values of $A^T A$ are the squares σ_i^2 of the singular values of A .

Hence forming $A^T A$ — as in the normal equations or the covariance matrix — **squares the condition number**, amplifying numerical errors for ill-conditioned matrices. **Direct SVD algorithms** compute the singular values of A without ever forming $A^T A$, working at condition number $\kappa(A)$ rather than $\kappa(A)^2$; they are therefore numerically preferable for ill-conditioned problems.